Bounds and Estimators for the Civilian-to-Combatant Death Ratio from Aggregation of Conflicting Sources

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Abstract

Casualty figures in armed conflict are reported by sources with opposing incentives: belligerents inflate enemy combatants killed and deflate civilians; advocacy and intergovernmental bodies push the other way. We treat the civilian-to-combatant death ratio as a partially identified parameter and ask what its value can be when conflicting sources are aggregated. The paper delivers two objects.

(1) Bounds from under-fudgible relations. Under a one-parameter model in which civilian adult men face μ times the death rate of women and children, we give a sharp identified set for the combatant share $q = D_M/D$ as a function of population structure, the demographic composition of the dead, and pre-war combatant stock; intersected with the mechanical bound $D_M \leq M$, this is a closed-form interval. Because the inputs are measured by different, non-aligned institutions, the relation is under-fudgible: moving any one input to rescue a disputed claim forces a compensating, visible move in an independently measured quantity. We define the *contradiction radius* ρ of a claim—the smallest standardised perturbation of an independent input that makes it feasible—turning "we dispute your number" into "name which measurement is wrong, and by how many standard errors."

(2) Confidence coefficients under political bias. Each source is allowed to be Huber ε_i-contaminated; the analyst's political-bias prior on a source enters as ε_i. We give a closed-form first-order sensitivity band that absorbs the assumed contamination, a patternwise worst-case envelope, and a quantile-displacement bound under an explicit posterior-weight condition.

In the Gaza 2023–2026 application—anchored on the Ministry of Health demographic record, whose composition is corroborated by an independent household survey—the identified set for q is $[0, 25\%]$ under only the weak assumption that civilian men are at least as exposed as women and children ($\mu \geq 1$), and $[0, 6\%]$ under exposure ratios calibrated on historical conflicts with known combatant status ($\mu \in [2, 3.5]$). The public Israel Defense Forces figure of 17–25k combatants killed converts to $q \approx 24$–36%. Its upper end is arithmetically infeasible: no exposure ratio $\mu \geq 1$ admits it, at a contradiction radius of 8–31 standard errors. Its lower end ($q \approx 24\%$) sits at the very edge of the $\mu \geq 1$ set—attainable only if civilian men died at essentially the same rate as women and children ($\mu \approx 1.04$)—and leaves the set under the slightest exposure differential, at a radius of 11 standard errors against the calibrated range. The rejection of the calibrated claim survives correcting the death toll upward and a 5% contamination allowance on every input. Independent evidence—demographic decompositions, the military's own leaked internal database of named militant dead, and a fully documented spatial Bayesian model—concentrates well below the claim, at civilian-to-combatant ratios of roughly 3:1 to 40:1 on direct deaths. The same machinery is symmetric: it rejects the polar claim of zero combatant deaths on documentary rather than demographic grounds.

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1 Introduction

1.1 The aggregation problem

Casualty figures in armed conflict are reported by sources with opposing incentives. Belligerents inflate enemy combatants killed and deflate civilians; advocacy organisations, opposing belligerents, and intergovernmental bodies have symmetric incentives in the other direction. Naive aggregation is therefore biased in a direction that depends on which side the analyst trusts more. The tempting conclusion is that civilian-to-combatant ratios are too politicised to analyse statistically.

We argue for a narrower position. Conflicting source aggregates fail in well-understood ways, but the parameters they purport to measure are jointly constrained by a small number of relations whose inputs are measured by different, non-aligned institutions. An analyst who treats every belligerent source as adversarial can still recover an informative bound by exploiting these constraints; what cannot be recovered without further assumptions is the position of the truth *within* the bound. This separation—bounds from data, position from politics—is the organising principle of the paper. The framework is a *diagnostic*, not an oracle: its purpose is to force any dispute about a casualty claim to name the specific independent measurement being contested, and to quantify how far that measurement would have to be wrong.

A worked example. Suppose a population is 73% women and children ($w = 0.733$) and records of the dead show that 56% of 70,000 dead are women and children ($\omega = 0.560$). If violence were blind to demographics, the dead would mirror the population, so the deficit of women and children among the dead is informative: some of the dead were combatants (all adult men, by assumption), and possibly civilian adult men were more exposed than their population share suggests. If civilian men and women die at *equal* per-capita rates, arithmetic (Section 3) turns these three numbers into a combatant share of $q \approx 25\%$. If civilian adult men die at *twice* the rate of women and children—the kind of differential found when the same method is calibrated on historical conflicts where the answer is known (Frost, 2026)—the same arithmetic gives $q \approx 6\%$. The interval $[\sim 6\%, \sim 25\%]$ is what the data alone deliver; a claimed q of 30% is outside it for *every* admissible exposure ratio and can only be rescued by asserting that the census or the death records are wrong, by an amount the framework computes in standard-error units. That amount is the claim's *contradiction radius*.

The closest precedents are Spagat et al. (2009) on the contestation of war-death estimates, Lum et al. (2013) on multiple systems estimation under reporting failure, Radford et al. (2023) on Bayesian inference of conflict losses with reporting biases, Vesco et al. (2026) on probabilistic underreporting in the Uppsala Conflict Data Program (UCDP), and Gómez-Ugarte et al. (2025) on uncertainty quantification for Gaza mortality. Methodologically we draw on partial identification (Manski, 2003, 2007; Molinari, 2020; Kline and Tamer, 2023) and robust Bayesian analysis (Huber, 1964, 1981; Berger and Berliner, 1986). On Gaza specifically, complementary demographic decompositions of the identified dead have been developed by Cockerill (2024) and Frost (2026), and we engage with both.

1.2 Two deliverables

Bounds. Where can the civilian-to-combatant ratio possibly lie, given a set of independently measured demographic and manpower facts? The answer is a sharp identified set in the sense of Manski's partial-identification programme (Manski, 2003; Tamer, 2010; Romano and Shaikh, 2010). If the identified set is narrow, no source-aggregation rule can pull the truth outside it. The width is determined by the data, not by the analyst's politics.

Confidence coefficients. How much can the analyst's political-bias prior on each source widen this set? Each source is Huber ε-contaminated (Huber, 1981; Berger and Berliner, 1986): it is true with probability $1 - \varepsilon$ and adversarial with probability ε. A bias-robust sensitivity band inflates the face-value interval by $\sum_i \varepsilon_i R_i$ to first order in ε, where R_i is the diameter of the identified set when source i is removed; conditional on a realised contamination of source i, the worst-case displacement is R_i itself. Setting $\varepsilon_i \approx 1$ for belligerent claims and $\varepsilon_i \approx 0.05$ for census-grade sources gives an interval honest about which inputs are doing the work.

The two objects are complementary: the first is information that survives any reasonable politics; the second is the rate at which uncertainty grows as the analyst dials in distrust of specific sources.

1.3 The under-fudgible mechanism

The bounds are not vacuous because of demographic arithmetic. Consider any claimed decomposition of the dead into combatants and civilians. The claim must simultaneously respect (i) the demographic composition of the identified dead, (ii) the demographic composition of the pre-war population, (iii) the size of the armed group's pre-war stock, and (iv) the plausible range of the civilian adult-male exposure differential. These quantities are measured by *different* institutions: a national or UN census produces (ii); a health ministry, the Office of the UN High Commissioner for Human Rights (OHCHR), or a household survey produces (i); independent military-balance assessments produce (iii); and (iv) can be calibrated on historical conflicts where combatant status was recorded (Frost, 2026). They are not equally manipulable, and—critically—they cannot all be moved at once without the moves being visible.

We call a relation between conflict statistics *under-fudgible* when its inputs are reported by non-aligned institutions (or have uncontroversial physical bounds), so that adjusting any single input to rescue a disputed claim forces a measurable deviation in at least one independently measured quantity. This is a heuristic property of a measurement system, not a formal axiom; what is formal is its consequence, the contradiction radius of Section 4, which quantifies exactly how large the forced deviation is for any specific claim. The trigger is *joint inconsistency*, not the size of the claim: the same machinery rejects a claimed combatant share that is too high and one that is too low, against the same feasible set. This symmetry matters. The diagnostic responds to the demographic content of a claim, not to its political direction, and Section 6 applies it in both directions.

1.4 Contributions and roadmap

The paper delivers, in order of importance:

(i) *A sharp identified set* for the combatant share $q = D_M/D$ under bounded differential exposure of civilian adult men, intersected with the manpower bound $D_M \leq M$ (Section 3). The set is a closed-form interval computable from four numbers.

(ii) *The contradiction radius* ρ: a feasibility test for any disputed claim, stated as a distance-to-feasibility optimisation in standard-error units (Section 4). It converts "we dispute your number" into "name which independent input is wrong, and by how many standard errors."

(iii) *A bias-robust sensitivity band* under Huber ε-contamination of each source: a pattern-wise worst-case envelope, a first-order band for the interval endpoints, and a quantile-displacement bound under an explicit posterior-weight condition (Section 5).

(iv) *A precision-weighted aggregation rule* for multiple demographic sources, with an explicit caveat for selection-biased sources (Section 2.1).

(v) *A Gaza 2023–2026 application* (Section 6): the identified set, the contradiction radius of the public Israel Defense Forces (IDF) combatant-death claim under every anchor and exposure assumption, a sensitivity grid over all contested inputs, and a comparison against every independent method known to us—which all reject the claim while disagreeing only within the identified set. A fully documented spatial Bayesian model (Appendix B) illustrates where a specific set of modelling assumptions lands *within* the set.

(vi) *An 81-conflict overview* (Section 7) mapping where the diagnostic can and cannot be applied sharply, with per-conflict tables in the supplement.

Proofs are collected in Appendix A. The method does not assign moral responsibility, estimate indirect deaths, or replace forensic work.

2 Setup

A conflict produces $D \in \mathbb{N}$ deaths in $[t_0, t_1]$ in a population of size N. Each death is a combatant ($C = 1$) or civilian ($C = 0$); write $D_M = \sum_i \mathbf{1}\{C_i{=}1\}$, $D_C = D - D_M$, and the targets

$$q = D_M/D \in [0, 1], \qquad r = D_C/D_M = (1 - q)/q.$$

Each death has a demographic class $k \in \mathcal{K} = \{\text{child}, \text{adult-female}, \text{adult-male}\}$. To keep the two central quantities visually distinct we reserve the *Latin* letter w for the population and the *Greek* letter ω for the dead:

$$\begin{aligned} w &= \text{women-and-children share of the } \underline{\text{population}}, \\ \omega &= \text{women-and-children share of the } \underline{\text{dead}}. \end{aligned}$$

Formally, $\omega_k = D_k/D$ is the share of class k among the dead and p_k its share in the population; AM denotes adult males (18 and over), with population share $a = p_{\text{AM}} = 1 - w$; $\omega = \sum_{k \neq \text{AM}} \omega_k$. Let M be the pre-war combatant stock and $f = M/N$. We take combatants to be in AM; female and minor combatants enter as a small slack $\bar{\eta} \leq 0.03$ (Remark 3).

Table 1. Core notation and measurement channels. The distinction between w (population) and ω (dead) is load-bearing throughout: w comes from a census taken before or independently of the fighting, ω from records of the dead.

Symbol	Meaning	Role	Typical source
D_M, D_C	combatant, civilian deaths	target	decomposition of D
$q = D_M/D$	combatant share of dead	estimand	—
D	total deaths in window	input	ministry of health, OCHA, UCDP, survey
w	women+children share *of population*	input	census, UN Population Division
$a = 1 - w$	adult-male (18+) share *of population*	input	census / age pyramid
ω	women+children share *of the dead*	input	identified-deaths records (e.g. a health ministry's)
M	combatant stock at t_0	input	IISS / RUSI / CSIS / intelligence
$[\mu_{\text{lo}}, \mu_{\text{hi}}]$	admissible exposure-ratio range	assumption	calibrated from historical conflicts
ρ	contradiction radius of a claim	output	standard-error units to feasibility; see §4

The last row anticipates the paper's diagnostic. The *contradiction radius* ρ of a disputed claim is the smallest number of measured standard errors by which at least one independent input (ω, w, or M) would have to move for the claim to become consistent with all the others; $\rho = 0$ means consistent, $\rho = 3$ means some independently measured quantity must be wrong by three standard errors, and values in the double digits mean the claim requires an input to be wrong by an amount that measurement error cannot plausibly produce. Section 4 makes this precise.

Sources. We use five channels: an independent population census (p_k, N); demographic records of the dead returning sample shares $\hat{\omega}_k$; aggregate death totals $\hat{D}$; belligerent combatant-death claims $\hat{D}_M$; and independent manpower estimates $\hat{M}$. Each source i is allowed to be ε_i-contaminated in Huber's sense (Huber, 1964; Berger and Berliner, 1986): it draws from the truth with probability $1 - \varepsilon_i$ and from an arbitrary adversary with probability ε_i.

2.1 Source aggregation

When several independent sources $i = 1, \ldots, m$ report the same scalar parameter θ with reported point estimates $\hat{\theta}_i$ and standard errors $\hat{\sigma}_i$, the natural *precision-weighted* aggregate is

$$\hat{\theta}_{\text{agg}} = \frac{\sum_i \hat{\theta}_i / \hat{\sigma}_i^2}{\sum_i 1/\hat{\sigma}_i^2}, \qquad \hat{\sigma}_{\text{agg}}^2 = \Big(\sum_i 1/\hat{\sigma}_i^2\Big)^{-1}. \tag{1}$$

Proposition 2.1 (Aggregation under conflicting sources)**.** *If sources are independent and unbiased, with variances treated as known (with estimated $\hat{\sigma}_i^2$ the statement holds conditionally on the variance estimates and asymptotically), $\hat{\theta}_{\text{agg}}$ in* (1) *is the minimum-variance unbiased linear estimator of θ. If source i has bias b_i, the aggregate has bias $(\sum_i b_i/\hat{\sigma}_i^2)/(\sum_i 1/\hat{\sigma}_i^2)$. Allowing ε_i-contamination (Section 5) in which the contaminated report is displaced from the truth by at most R_i, the worst-case bias of $\hat{\theta}_{\text{agg}}$ is at most $\sum_i \varepsilon_i R_i \cdot \hat{\sigma}_{\text{agg}}^2 / \hat{\sigma}_i^2$.*

Proof in Appendix A.

Precision weighting presupposes that the sources measure the *same* estimand. When one source is subject to a known selection mechanism—as with the OHCHR verification sample in the Gaza application, which is heavily weighted toward residential-strike deaths (Spagat and Cockerill, 2024)—the honest treatment is not to blend it into the aggregate but to use it as a stratum-specific measurement and anchor the analysis on the source whose selection is best understood. Section 6 applies this rule: the Ministry of Health record, whose demographic composition is corroborated by an independent household survey (Spagat et al., 2026a), is the primary anchor, and results are reported for the alternative anchors so the reader can see exactly how anchor choice propagates.

3 Bounds: partial identification of q

This section delivers a sharp identified set for q that any source-aggregation rule must respect. We begin with a benchmark assumption, then weaken it. Proofs are in Appendix A.

Assumption 3.1 (Exchangeable collateral)**.** Civilians of all demographic classes face equal per-capita death hazard.

Under Assumption 3.1 the combatant share is point-identified by a single rearrangement: civilian deaths split between the women-and-children class and the adult-male class in proportion to their civilian population shares, so

$$q = 1 - \frac{\omega(1-f)}{w}. \tag{2}$$

The intuition is direct: ω falls short of the population share w only to the extent that deaths are diverted to the (all-male) combatant pool.

Remark. Suppose a fraction $\eta \leq \bar{\eta} = 0.03$ of combatant *deaths* is recorded in class W. This is a classification slack: the combatant stock itself stays in AM, so the civilian split $S = w/(w + \mu(a - f))$ is unchanged. The identity becomes $\omega = (1 - q)S + \eta q$, the true share is $q_{\text{true}} = (S - \omega)/(S - \eta)$, and the no-slack value $q_{\eta=0} = (S - \omega)/S$ understates it by exactly

$$q_{\text{true}} - q_{\eta=0} \;=\; \frac{\eta\, q_{\text{true}}}{S} \;=\; \frac{\eta\, q_{\eta=0}}{S - \eta} \;\geq\; 0 \qquad (S > \eta,\; q_{\text{true}} \geq 0).$$

At the values in Section 6, where $S > \bar{\eta}$ throughout ($S \geq 0.597$ for $\mu \leq 2$) and $q_{\eta=0} \leq 0.26$, a loose uniform bound over $\mu \in [1, 2]$ is $\bar{\eta} \cdot 0.26/(0.597 - \bar{\eta}) \approx 1.4$ percentage points; the exact understatement is smaller and falls with μ (1.05 points at $\mu = 1$, 0.33 at $\mu = 2$), since $q_{\eta=0}$ shrinks faster than S. Readers who want this slack should widen the *upper* endpoints below by $\bar{\eta}\, q_{\eta=0}/(S - \bar{\eta})$; we otherwise suppress it.

Assumption 3.1 is strong: civilian adult men typically face higher per-capita exposure than women and children due to outside work, food queues, rescue activity, and circumstantial proximity to combat. We weaken it to a one-parameter family.

Assumption 3.2 (μ-bounded differential exposure)**.** There exists $\mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$ such that, conditional on being civilian, class AM has per-capita death rate μ times that of class W.

Theorem 3.3 (Sharp identified set)**.** *Suppose $f \leq a$ and Assumption 3.2 holds. Given (ω, w, a, f) and the exposure range $[\mu_{\text{lo}}, \mu_{\text{hi}}]$, the identified set for q is the image of the monotone-decreasing curve*

$$q(\mu) = 1 - \omega\, \frac{w + \mu(a - f)}{w}, \qquad \mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}], \tag{3}$$

giving the sharp interval $[q(\mu_{\text{hi}}),\, q(\mu_{\text{lo}})] \cap [0, 1]$.

Corollary 3.4 (Manpower bound)**.** *$q \leq M/D$, so whenever the endpoints below are ordered, the joint feasible interval is*

$$q \in \big[\max\{0,\, q(\mu_{\text{hi}})\},\; \min\{1,\, q(\mu_{\text{lo}}),\, M/D\}\big];$$

if the lower endpoint exceeds the upper the set is empty and the inputs are jointly infeasible—the event detected by a positive contradiction radius in Section 4.

Calibrating the exposure range. The exposure ratio μ is the framework's only behavioural parameter, and it can be disciplined empirically rather than stipulated. Frost (2026) identifies the civilian male exposure differential from the male-to-female death ratio just *above* the combatant age range—where the combatant share is mechanically near zero—and validates the procedure on historical Gaza conflicts recorded by B'tselem, where combatant status is known: there, the observed male-to-female ratio at the top of the combatant age band is 3.4:1 while the effective differential through the combatant ages is 5:1, so the observable ratio is itself a conservative lower bound. Airstrike-specific data suggest smaller differentials (≈ 1.3; Cockerill, 2024), consistent with indiscriminate mechanisms; ground operations push the differential up. We therefore report the identified set on a grid $\mu \in \{1, 1.5, 2, 2.5, 3.5, 5\}$ throughout, treating $[2, 3.5]$ as the empirically best-supported range for a mixed air-and-ground campaign and the extreme values as stress tests. Larger μ *lowers* the implied combatant share; the reader who disputes a high-μ calibration is pushed toward *higher* q, and the sensitivity grid (Table 3) makes the entire mapping explicit.

Plug-in estimator and standard error. The sample analogue of the upper endpoint is $\hat{q} = 1 - \hat{\omega}(1 - \hat{f})/\hat{w}$, with delta-method variance

$$\widehat{\text{Var}}(\hat{q}) \approx \frac{\hat{\omega}^2}{\hat{w}^2}\widehat{\text{Var}}(\hat{f}) + \frac{(1 - \hat{f})^2}{\hat{w}^2}\widehat{\text{Var}}(\hat{\omega}) + \frac{\hat{\omega}^2(1 - \hat{f})^2}{\hat{w}^4}\widehat{\text{Var}}(\hat{w}).$$

4 Bounds: contradiction radius

Theorem 3.3 returns an interval; a public claim returns a point. The diagnostic measures their distance after standardising each input by its measurement uncertainty.

Definition 4.1 (Feasible set)**.** For inputs $x = (\omega, w, a, f, M, D)$ and exposure range $[\mu_{\text{lo}}, \mu_{\text{hi}}]$,

$$\mathcal{F}(x) = \Big\{ q \in [0,1] : \exists\, \mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}] \text{ with } q = 1 - \omega \tfrac{w+\mu(a-f)}{w} \text{ and } qD \le M \Big\}. \tag{4}$$

A claim $\hat{D}_M^{(i)}$ converts to $q^{(i)} = \hat{D}_M^{(i)}/\hat{D}$ and is feasible iff $q^{(i)} \in \mathcal{F}(\hat{x})$.

Definition 4.2 (Contradiction radius)**.** Let $\hat{x} = (\hat{\omega}, \hat{w}, \hat{a}, \hat{f}, \hat{M}, \hat{D})$ with measured standard errors $\sigma_\omega, \sigma_w, \sigma_M$. The contradiction radius of claim $q^{(i)}$ is

$$\rho^{(i)} = \inf_{\omega', w', M'} \max\Big\{ \tfrac{|\omega' - \hat{\omega}|}{\sigma_\omega},\ \tfrac{|w' - \hat{w}|}{\sigma_w},\ \tfrac{(M' - \hat{M})_+}{\sigma_M} \Big\} \quad \text{s.t.} \quad q^{(i)} \in \mathcal{F}((\omega', w', \hat{a}, \hat{f}, M', \hat{D})). \tag{5}$$

Theorem 4.3 (Feasibility and contradiction)**.** *With $(\hat{a}, \hat{f}, \hat{D})$ and the exposure range fixed, a claim is consistent with the independent inputs iff $\rho^{(i)} = 0$. If $\rho^{(i)} > 0$, every rationalisation (ω', w', M') makes at least one* penalised *displacement—$|\omega' - \hat{\omega}|/\sigma_\omega$, $|w' - \hat{w}|/\sigma_w$, or $(M' - \hat{M})_+/\sigma_M$—of at least $\rho^{(i)}$ standard-error units. (Lowering M is unpenalised because it can only shrink the feasible set.)*

Interpretation. The contradiction radius answers the question a motivated critic must eventually face: *which measurement, exactly, do you say is wrong, and by how much?* A claim with $\rho = 0$ is compatible with all independent inputs; nothing more can be said against it on this evidence. A claim with $\rho = 2$ requires some input to be off by two standard errors—uncomfortable but conceivable. A claim with $\rho = 20$ requires an independently measured quantity to be wrong by twenty standard errors, an error that measurement noise cannot produce; defending such a claim requires asserting institutional fabrication, and the radius names the institutions and the magnitudes. Because ρ standardises by *measured* uncertainty, it is unit-free and comparable across conflicts and across claims. It is also symmetric: an implausibly *low* combatant claim generates a positive radius the same way an implausibly high one does. What the radius is not: it is not a posterior probability, it does not weigh how *many* inputs must move (only the largest single standardised move), and it inherits any systematic—as opposed to sampling—error in the inputs, which is why Section 6 reports it under every anchor choice.

Computation. Because $q(\mu)$ is monotone, the boundary of $\mathcal{F}$ is attained on one of three axes: the demographic axis (ω moves), the population-composition axis (w), or the manpower axis (M/D). This is the contradiction triangle of Figure 1; (5) is a low-dimensional convex programme.

Diagnostic procedure.

D1. Fix the analysis window and total deaths D; carry the uncertainty in D as a sensitivity dimension, not a constant.

D2. Estimate (w, a) from the population baseline and ω from identified-deaths records, checking each candidate ω source for selection mechanisms before use.

D3. Calibrate the exposure range $[\mu_{\text{lo}}, \mu_{\text{hi}}]$ from historical conflicts with known combatant status, and take an independent upper bound on M.

D4. Compute $\mathcal{F}(\hat{x})$ from (4).

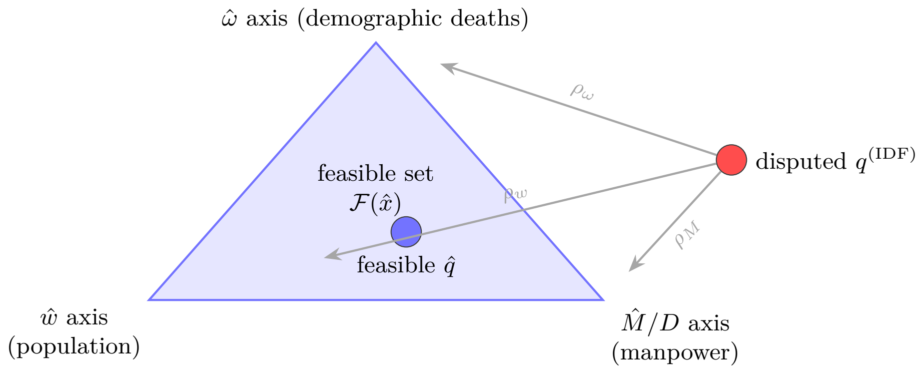

Figure 1. Contradiction triangle (conceptual schematic). The feasible set $\mathcal{F}$ is bounded by three independently measured "axes"; the contradiction radius $\rho = \min(\rho_\omega, \rho_w, \rho_M)$ is the shortest standardised perturbation from a disputed claim into $\mathcal{F}$. Each axis is owned by a different institution, which is what makes the relation under-fudgible: rescuing a claim on one axis leaves fingerprints on that axis's measurement.

D5. For every disputed $\hat{D}_M^{(i)}$, report $q^{(i)} = \hat{D}_M^{(i)}/D$, the membership $q^{(i)} \in \mathcal{F}$, and $\rho^{(i)}$, under each anchor and each D scenario.

The test is symmetric: it rejects "too many combatants" and "zero combatants" against the same set.

5 Confidence coefficients under political bias

The bounds in Sections 3–4 treat each measured input as honest. We now allow the analyst to dial in distrust of specific sources. Political bias enters the confidence statement through Huber contamination: each source i writes its likelihood as $(1-\varepsilon_i)L_i + \varepsilon_i Q_i$, with Q_i adversarial in some class $\mathcal{Q}$ (Huber, 1964, 1981; Berger and Berliner, 1986; Gagnon, 2023). The contamination parameter $\varepsilon_i \in [0,1]$ is the analyst's prior probability that source i is producing a politically motivated reading rather than a measurement of the true distribution.

Proposition 5.1 (Bias-robust sensitivity band)**.** *Let θ be a scalar estimand with $1-\alpha$ posterior credible interval $I_0 = [L_0, U_0]$ under face-value sources, with the posterior supported on the identified set implied by the sources used. Suppose each source i is independently contaminated with probability ε_i, and removing the sources in a set T leaves an identified set of diameter R_T (write $R_i := R_{\{i\}}$).*

(i) (Patternwise envelope.) *If the realised contamination pattern is T, every* support-respecting $1-\alpha$ *credible interval—one whose endpoints lie in the posterior support, e.g. the equal-tailed or HPD interval—obtainable by adversarial replacement of the sources in T lies within $[L_0 - R_T,\ U_0 + R_T]$. (An arbitrary credible interval can be widened without limit while retaining $1-\alpha$ coverage, so no finite envelope constrains that class.)*

(ii) (Expected displacement.) *Averaging over the contamination pattern, the expected endpoint displacement is at most $\sum_i \varepsilon_i R_i$ plus higher-order terms in ε involving the multi-removal diameters R_T, $|T| \geq 2$; when at most one source can be contaminated, the bound $\sum_i \varepsilon_i R_i$ is exact. The band*

$$I_\varepsilon \;=\; \Big[L_0 - \textstyle\sum_i \varepsilon_i R_i,\ U_0 + \sum_i \varepsilon_i R_i\Big] \tag{6}$$

is therefore a first-order sensitivity band for the interval endpoints—not a uniform envelope over realised contaminated intervals, whose endpoints can move by up to R_T on the (probability $\leq \varepsilon_i$) contamination events.

(iii) (Quantile displacement.) *If additionally the posterior mixture weight on contaminated components is at most $\tau := 1 - \prod_i (1-\varepsilon_i)$—which holds when the component marginal likelihoods are comparable, and must otherwise be checked, since posterior weights are proportional to prior pattern probabilities* times *component marginal likelihoods—and the face-value posterior density exceeds $m > 0$ on the segment between the clean and contaminated u-quantiles, then*

$$\left|\hat{Q}_\varepsilon(u) - \hat{Q}_0(u)\right| \;\le\; m^{-1}\tau, \qquad u \in \{\alpha/2,\, 1-\alpha/2\}.$$

Proof in Appendix A.

Practice. Set $\varepsilon_i \approx 0.05$ for widely-trusted demographic sources (Palestinian Central Bureau of Statistics, health-ministry demographic records corroborated by independent surveys) and $\varepsilon_i \approx 1$ (i.e., adversarial) for belligerent claims; (6) then absorbs the assumed political bias automatically. In the Gaza application the removal diameters are $R_{\text{demo}} = 0.64$ (dropping the demographic records leaves only the manpower bound $q \le M/D$), $R_{\text{census}} = 0.06$ (letting w range over a generous global envelope $[0.70, 0.76]$), and $R_M = 0$ (the manpower bound never binds), so $\varepsilon = 0.05$ inflates the exposure-agnostic upper bound by 3.5 percentage points—from 25.1% to 28.7%—which still rejects the upper end of the disputed claim.

6 Application: Gaza 2023–2026

We instantiate the framework using only inputs whose measurement is not controlled by either belligerent: Palestinian Central Bureau of Statistics (PCBS) population structure (Palestinian Central Bureau of Statistics, 2023–2024); total deaths from the UN Office for the Coordination of Humanitarian Affairs (OCHA) and the Gaza Ministry of Health (MoH) (United Nations Office for the Coordination of Humanitarian Affairs, 2025–2026); OHCHR and MoH demographic death records (United Nations Office of the High Commissioner for Human Rights, 2024); the Gaza Mortality Survey (GMS) household study (Spagat et al., 2026a); excess-mortality information (Khatib et al., 2024, 2025); and combatant-strength estimates from the International Institute for Strategic Studies (IISS), the Royal United Services Institute (RUSI), and the Center for Strategic and International Studies (CSIS) (International Institute for Strategic Studies, 2024; Royal United Services Institute, 2024). The IDF claim itself (Israel Defense Forces, 2024–2026) is the object of the test, not an input. All numbers in this section are produced by a validation script in the companion repository that recomputes every bound, radius, and table entry from the raw inputs and fails if any quoted figure disagrees.

Anchor choice. The two demographic records of the dead measure different things. The OHCHR sample ($n = 8{,}119$) applies a strict three-source verification rule under which 7,607 of the 8,119 verified deaths (94%) occurred in residential buildings, where women and children predominate (Spagat and Cockerill, 2024); it is best read as a stratum-specific measurement, not an estimate of the overall ω. The MoH full record ($n \approx 70{,}000$) covers all recorded deaths, and its demographic composition is independently corroborated by the Gaza Mortality Survey, a household probability survey that also quantifies the MoH undercount at roughly 35% (Spagat et al., 2026a); the Israeli military has itself since accepted the MoH cumulative toll (Kubovich and Hasson, 2026). We therefore anchor on the MoH record ($\omega = 0.560$, conservative $\sigma_\omega = 0.01$) and report the OHCHR anchor—and a geometric blend of the two—as sensitivities rather than blending them into a single aggregate. Because $\omega_{\text{MoH}} < \omega_{\text{OHCHR}}$, this choice is *favourable* to high-combatant claims: every rejection below survives, a fortiori, under the alternative anchors.

The survey's representativeness has been contested: DellaPergola and Zlochin (2026) identify interviewer-level heterogeneity and protocol deviations and argue the population-level

Table 2. Diagnostic inputs for Gaza. The disputed quantity D_M is tested against the other rows. The adult-male class is males 18 and over, so $a = 1 - w$ exactly.

Input	Value	Channel
w (women+children *of population*)	0.733	PCBS population structure
a (males 18+ *of population*)	0.267	$= 1 - w$
ω (women+children *of the dead*)	0.560 (MoH record; primary anchor)	identified deaths, corroborated by GMS
	0.693 (OHCHR verified sample)	residential-strike stratum
D (direct deaths, late 2025)	70k (MoH) / 94k (GMS-corrected) / 106k (+missing)	MoH, GMS, missing-persons registry
f (combatant share of population)	0.0157–0.0269 (0.020 central)	Hamas / Palestinian Islamic Jihad pre-war manpower / population
μ (exposure ratio)	grid $\{1, 1.5, 2, 2.5, 3.5, 5\}$; calibrated $[2, 3.5]$	Frost (2026); Cockerill (2024)
M (combatant strength)	$\leq$ 45,000	IISS / RUSI / CSIS
claim (tested, not an input)	17–25k combatants killed	IDF public statements

extrapolation is unreliable; Spagat et al. (2026b) respond that the conclusions survive excluding the contested teams (the violent-death estimate falls from 75,200 to 59,200 under maximal exclusion, with the demographic composition intact). For present purposes the dispute is instructive precisely because it does not touch the bounds. It concerns the *level* of the death toll, an input our identified set does not use: $q(\mu)$ depends on (ω, w, a, f) only, and the demographic composition—the one quantity the framework takes from the survey—is stable across every specification in the exchange, including the critics' own exclusions and the extreme all-missing-dead scenario (the survey's women-children-elderly share of violent deaths moves from 56.2% only to 52.2%; note this classification includes people over 64 and so is not identical to our ω class, but it is the composition measure the exchange contests). The undercount factor enters solely through the D-sensitivity dimension of Step 3 ($D = 70$–106k), and only its upward corrections matter there. Downward revisions of the toll require no new scenario, because the direction of the challenge is self-defeating for the claim it might appear to assist: if the critics were right and the true toll were *lower*, the claimed combatant counts would convert to *higher* shares $q^{(i)} = \hat{D}_M^{(i)}/D$, moving the disputed claim further outside the feasible set while leaving the set itself unchanged. This is the under-fudgible structure at work: contesting one measurement re-routes, but does not relieve, the burden of the contradiction.

Step 1: rejecting uniform random violence. The naive null "violence is uniform across the population" predicts $\mathbb{E}\,\omega = w = 73.3\%$. Even the OHCHR sample ($\hat{\omega} = 69.3\%$) rejects this at $z = -8.1$ ($p \approx 1.9 \times 10^{-16}$); the MoH record ($\hat{\omega} = 56.0\%$) rejects it overwhelmingly. Adult males are overrepresented among the dead—necessarily, since the universe contains both combatants and structurally more-exposed civilian men. Rejecting uniformity is necessary but not sufficient to rationalise any *specific* combatant share; the question is how the adult-male excess splits between the two explanations, and that is precisely what μ indexes.

Step 2: the identified set. On the MoH anchor, Theorem 3.3 gives

$$q(1) = 25.1\%, \qquad q(1.5) = 15.7\%, \qquad q(2) = 6.3\%, \qquad q(\mu) \leq 0 \text{ for } \mu \geq 2.3.$$

The exposure-agnostic set (any $\mu \geq 1$; the upper endpoint is $q(1)$ and the lower is clipped at zero) is $q \in [0,\ 25.1\%]$; the empirically calibrated set ($\mu \in [2, 3.5]$, Frost, 2026) is $q \in [0,\ 6.3\%]$. The manpower bound $M/D = 0.64$ never binds. Table 3 reports the full anchor$\times\mu$ grid; the OHCHR anchor caps q at 7.3% already at $\mu = 1$.

Table 3. Sensitivity of the identified-set upper endpoint. Each cell is $q(\mu) = 1 - \omega\,(w + \mu(a - f))/w$ evaluated at $w = 0.733$, $a = 0.267$, $f = 0.020$: the largest combatant share consistent with anchor ω if civilian adult men die at exactly μ times the per-capita rate of women and children. The identified set for exposure range $[\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}]$ is $[q(\mu_{\mathrm{hi}}),\ q(\mu_{\mathrm{lo}})] \cap [0, 1]$. Dashes: $q(\mu) \leq 0$, i.e. the anchor is consistent with zero combatants at that exposure ratio. Empirically calibrated exposure ratios are $\mu \gtrsim 2$ (Frost, 2026; Cockerill, 2024).

Anchor ω (share of dead)	μ=1	μ=1.5	μ=2	μ=2.5	μ=3.5	μ=5
MoH (GMS-validated), $\omega = 0.560$	25.1%	15.7%	6.3%	–	–	–
Blend (geometric), $\omega = 0.620$	17.1%	6.7%	–	–	–	–
OHCHR (residential-strike stratum), $\omega = 0.693$	7.3%	–	–	–	–	–

Step 3: testing the IDF claim. Over the MoH-confirmed denominator $D = 70\text{k}$, the claimed 17–25k combatants convert to $q^{\mathrm{IDF}} = 24.3\%–35.7\%$, i.e. a civilian-to-combatant ratio of roughly 2:1–3:1. The contradiction radius of Definition 4.2—which grants the claim the most favourable μ in the admissible range—splits cleanly across the two endpoints:

- *The* 25*k endpoint is infeasible outright.* Rationalising $q = 35.7\%$ on the MoH anchor requires $\mu = 0.44$: civilian adult men would have to die at less than half the per-capita rate of women and children, which no mechanism in this conflict supports. On the demographic axis the radius is $\rho_\omega \approx 7.9$ standard errors with agnostic exposure and ≈ 17.6 with calibrated exposure (and $\approx 21–31$ on the OHCHR anchor).
- *The* 17*k endpoint is feasible only at the edge, and only without calibration.* At $q = 24.3\%$ the required exposure ratio is $\mu = 1.04$: barely inside the mathematically admissible range, so $\rho = 0$ on the MoH anchor only if civilian men in Gaza were essentially no more exposed than women and children. Any exposure differential at all puts the claim outside the set; the empirically calibrated range $\mu \in [2, 3.5]$—obtained from historical Gaza conflicts where combatant status is recorded (Frost, 2026)—puts it $\rho_\omega \approx 10.8$ standard errors outside. The lower endpoint is therefore feasible only marginally, resting on the single most claim-favourable value of the one parameter the data cannot pin down. And even that value buys only the bottom of the claimed range: $\mu = 1$ caps the combatant share at 25.1%, roughly 17.6k of the 70k dead, so every count the IDF asserts above that—most of the 17–25k interval—requires, *on top of* $\mu = 1$, moving the census w or the demographic record ω as well. Granting equal exposure is thus not enough to rationalise the claim; it only relocates the remaining burden onto quantities other institutions measure.

The rejection is robust to the denominator: correcting the MoH undercount ($D = 94\text{k}$) and adding the missing ($D = 106\text{k}$) lowers q^{IDF} to 16–24%, but the required exposure ratios ($\mu \leq 1.5$ for 17k, $\mu \leq 1.1$ for 25k) remain below the calibrated range. (Applying the recorded ω to the corrected totals assumes the unrecorded dead share the recorded demographic mix, which the survey's demographic breakdown supports.) The first-order sensitivity band of Proposition 5.1, with $\varepsilon = 0.05$ on the census and demographic records, inflates the exposure-agnostic upper endpoint from 25.1% to 28.7%—still below the claim's upper endpoint. (The patternwise worst case is starker but works in the same direction: if the demographic record itself is the contaminated source, the analysis reduces to the manpower bound $q \leq 0.64$, which is exactly why the record's corroboration by an independent household survey carries the weight it does.)

Step 4: the named-combatant cross-check. A joint investigation by *The Guardian*, +972 Magazine, and Local Call (Kierszenbaum et al., 2025) reported that the Israeli Military Intelligence directorate's internal database listed 8,900 named Hamas and Palestinian Islamic Jihad operatives dead or probably dead as of May 2025, against a contemporaneous recorded toll of ∼53,000: $q \approx 16.8\%$, inside the exposure-agnostic identified set and far below the public claim of the same period. We treat this as corroborating, not primary, evidence—the database's coverage and classification rules are not public—but it is notable that the belligerent's own accounting apparatus lands inside the bound its public statements violate.

The manpower ledger. The claim's coherence can also be audited entirely within the belligerent's own accounting, using no Palestinian-measured input at all. This audit does not move the bounds; it prices the remaining escape route. The IDF's official pre-war estimate of Hamas strength is 30,000 fighters in 24 battalions (The Jerusalem Post, 2024). Its assessment after the October 2025 ceasefire is that more than 22,000 operatives were killed—and that the military wing nevertheless fields roughly 20,000 operatives today (Meir Amit Intelligence and Terrorism Information Center, 2026). These figures are jointly consistent only if at least 12,000 operatives were recruited during the war and the ceasefire, before counting the wounded and the detained; United States intelligence reported recruitment on exactly this scale (10–15k by January 2025, roughly matching losses over the same period; Reuters, 2025). The ledger does not say where the recruits sit, but both placements concede ground. If they are among the dead, the killed-operatives count includes people who were civilians when the pre-war stock was measured, so the count stops being an answer to the question the public claim poses—how much of the force that existed on 7 October was destroyed. If they are among the living, the killed may all be pre-war stock, but the claimant then asserts that recruitment ran at the pace of attrition—the wartime-recruitment concession of Section 8, made by its own assessment rather than by its critics. Either way, rescuing the claim on the manpower axis costs exactly what the framework says it must.

Step 5: convergence of independent methods. Table 4 is the central empirical exhibit. Methods with different data, assumptions, and failure modes—the identified set of this paper, demographic decompositions (Cockerill, 2024; Frost, 2026), the military's internal database, and a fully specified spatial Bayesian model (Appendix B)—all concentrate in $q \in [\sim 2\%, \sim 25\%]$, civilian-to-combatant ratios from roughly 3:1 to 40:1 or more on direct deaths. The upper end of the public claim ($q = 35.7\%$) lies outside this range for all of them; the lower end ($q = 24.3\%$) touches only the mathematical edge of the widest ($\mu \geq 1$) bound and falls outside every method that admits any exposure differential. The spatial model, whose priors and structure are documented in Appendix B, posterior-concentrates at $q = 2.5\%$ (95% credible interval $[1.6\%, 3.8\%]$, civilian-to-combatant ratio 43:1 $[30, 64]$); we report it as one model-based point within the identified set, not as the headline, because its position within the set is prior-sensitive in ways the bounds are not.

Symmetry. The same machinery tests the polar narrative $q = 0$ ("all the dead are civilians"). On the MoH anchor, $q = 0$ requires $\mu = 2.33$—inside the calibrated range—so the demographic axis alone cannot reject it; the arithmetic is compatible with a purely civilian death toll if civilian men were about 2.3 times as exposed. What rejects $q = 0$ is the named-combatant evidence: thousands of individually named militant dead are documented in the military's own

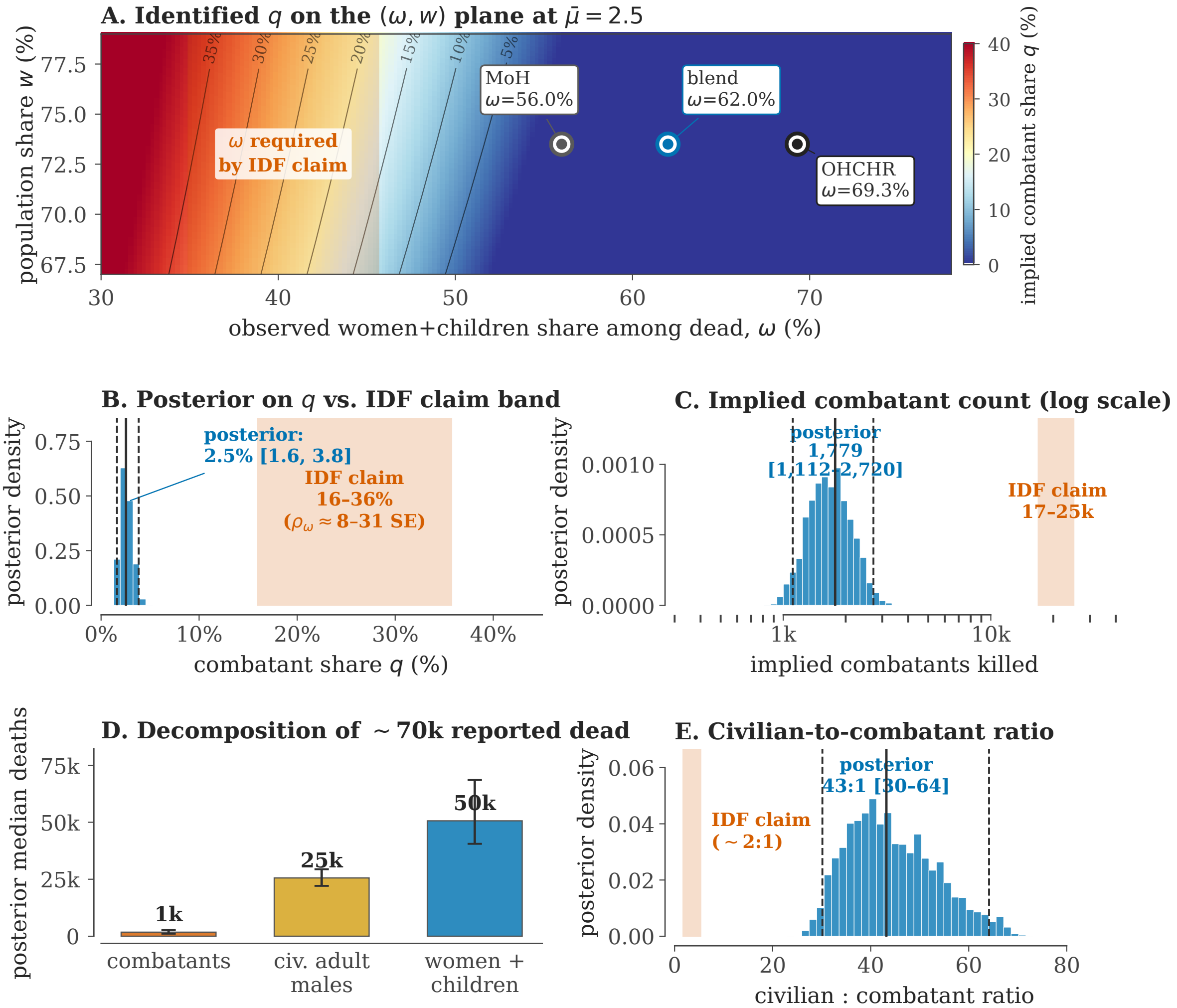

Figure 2. The diagnostic for Gaza (2023–2026). **(A)** Identified q on the (ω, w) plane at $\mu = 2.5$. Iso-q contours are black; the OHCHR, blended, and Gaza Ministry of Health (MoH) anchors all sit in the low-q (blue) region; the IDF claim requires ω in the red band on the far left, far from any observed anchor. **(B)** Histogram of the spatial posterior on q (median, 95% credible interval as solid+dashed lines) vs. the IDF claim band converted to q. **(C)** Implied combatants killed (log axis) vs. the IDF claim of 17–25k. **(D)** Posterior decomposition of the ~70k reported dead under the spatial model: ~1–3k combatants, ~25k civilian adult men, ~45–50k women and children. **(E)** Civilian-to-combatant ratio: spatial-model posterior median 43:1 vs. the IDF-implied ~ 2:1. Panels B–E display the spatial model of Appendix B; the paper's headline claims rest on panels A and Table 4, not on the posterior.

Table 4. Convergence of independent methods on the Gaza combatant share. Every method concentrates in $q \in [\sim 2\%, \sim 25\%]$ (civilian-to-combatant ratios of roughly 3:1 to 40:1 or more on direct deaths). The upper end of the IDF claim (35.7%) lies above all of them; its lower end (24.3%) only touches the mathematical edge of the widest ($\mu \geq 1$) bound and lies above every method that admits any exposure differential.

Method	Source	Implied q	Note
IDF public claim (17–25k of 70k)	(Israel Defense Forces, 2024–2026)	24.3–35.7%	object of the test
Identified set, MoH anchor, $\mu \geq 1$	this paper	0.0–25.1%	weakest assumption ($\mu \geq 1$)
Identified set, MoH anchor, $\mu \in [2, 3.5]$	this paper	0.0–6.3%	Frost-calibrated exposure
Aman named-militant database	(Kierszenbaum et al., 2025)	16.8%	8,900 named dead of 53,000 (May 2025)
Male-bias model, MoH record	(Frost, 2026)	11.1–16.9%	B'tselem-calibrated
Demographic decomposition	(Cockerill, 2024)	9.4–26.3%	range of male-bias assumptions
Spatial Bayesian posterior	this paper, App. B	1.6–3.8%	prior-dependent; one point in the set

database (Kierszenbaum et al., 2025). The diagnostic thus does exactly what it should: it rejects the high claim on demographic grounds, rejects the zero claim on documentary grounds, and reports which axis does the work in each case.

7 Empirical overview: 81 conflicts, 1900–2026

The contradiction-radius diagnostic requires demographic micro-data and manpower estimates, and cannot be applied sharply to every historical conflict. The 81-conflict dataset is used here only to map where civilian-share uncertainty is large, where indirect mortality dominates, and which conflicts would benefit most from identified-deaths samples; the supplement contains the per-conflict tables.

Generalisation. The diagnostic transfers to any conflict with (i) a population demographic decomposition, (ii) an independent demographic survey of the dead, (iii) an independent manpower estimate at t_0, and (iv) a reported aggregate death total. Such inputs exist for Tigray 2020–2022 (Ethiopian Statistical Service plus WHO surveys), Syria 2011– (Violations Documentation Center plus Syrian Network for Human Rights), the Mexican drug war (INEGI, the national statistics institute, plus CSIS/IISS cartel estimates), the Iraq War 2003–2011 (Burnham et al., 2006), and Yemen 2014– (Yemen Data Project plus the Armed Conflict Location and Event Data project, Raleigh et al., 2010). Where these data are missing, the supplement marks the gap rather than inventing a contradiction radius.

8 Discussion

What the paper claims, and what it does not. The conceptual contribution is the contradiction triangle (Theorem 4.3): no single number can be manipulated without forcing matching deviations in independently-measured quantities, and the contradiction radius ρ gives a unit-free measure of how far a claim sits from the consensus-feasible set. Whereas Spagat

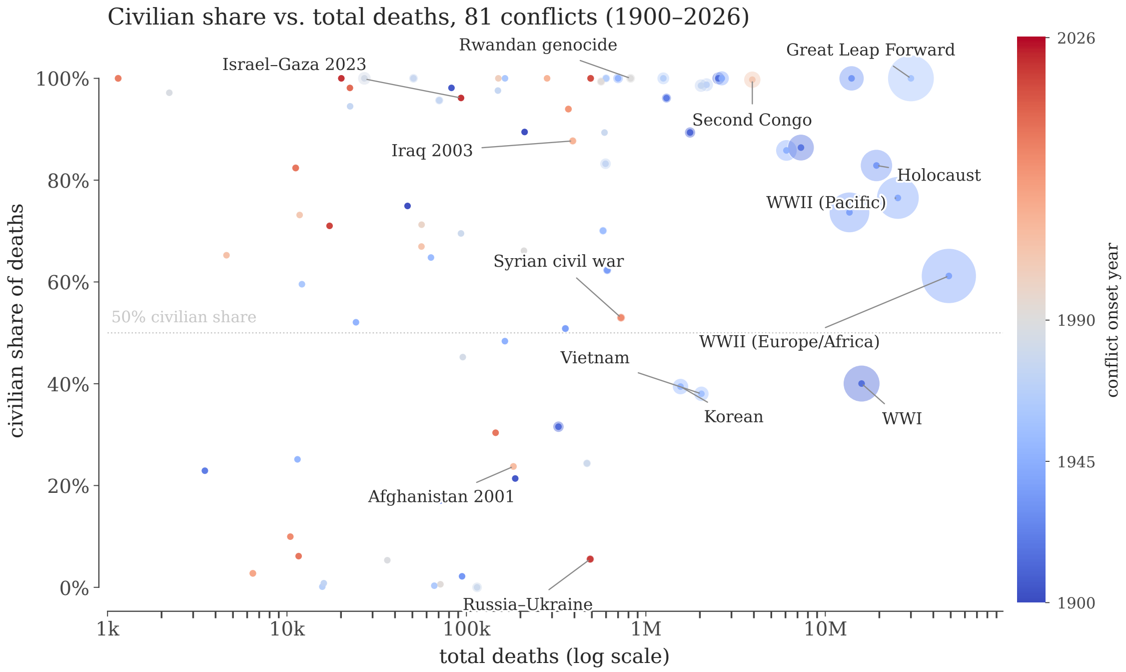

Figure 3. Civilian share vs scale, 1900–2026. Markers are conflicts; area is proportional to midpoint total deaths. Wars cluster bimodally: combat-dominated interstate wars (civilian share $\lesssim 30\%$) and civilian-targeting events, famines, and atrocities (civilian share $\gtrsim 90\%$).

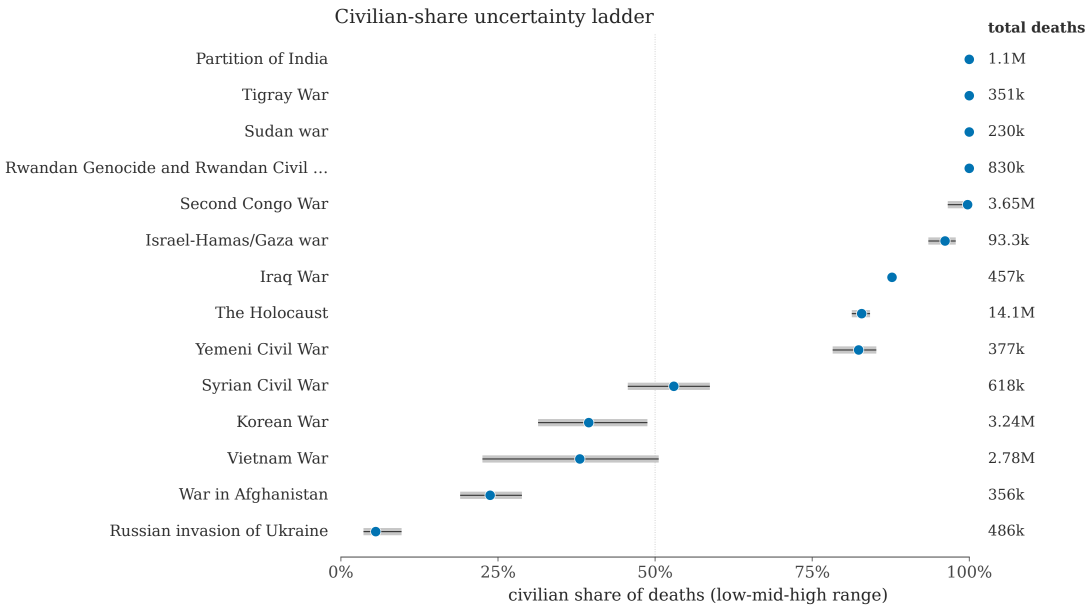

Figure 4. Civilian-share uncertainty ladder. Selected high-leverage conflicts. Grey segments are the side-level low/high implied civilian-share interval; blue dots are the midpoints. Wide intervals mark conflicts where the next technical advance is a better identified-deaths sample, not a better plot.

et al. (2009) frame war-death estimation as an *arena of contestation*, our framework gives the analyst a numerical referee for that arena, in the spirit of Lum et al. (2013) for multiple systems estimation and Radford et al. (2023) for Bayesian source aggregation. In the Gaza application the referee's ruling is deliberately modest: the data bound the combatant share of direct deaths at 25% with no behavioural assumptions and at about 6% under empirically calibrated exposure, and the public claim of 24–36% is rejected under every anchor and every calibrated exposure assumption we examined—its upper endpoint is arithmetically infeasible outright, and its lower endpoint survives only by asserting that civilian men were no more exposed than women and children ($\mu \approx 1$), against the calibration evidence. The paper does *not* claim that the spatial model's $q = 2.5\%$ is the truth; that number is one prior-dependent point inside the set, reported with its full model documentation so that readers can locate exactly which assumptions produce it.

Where the power comes from. It is worth being explicit about why the bound binds. The demographic relation has one free behavioural parameter, the exposure ratio μ, and the data pin the product, not the factors: a given deficit of women and children among the dead can be explained by combatants dying, by civilian men being more exposed, or by a mixture. High-combatant claims must therefore assert that $\mu \approx 1$—that civilian men in an urban war were no more exposed than women and children—at the same time as independent calibrations on conflicts with recorded combatant status put μ at 2 or higher (Frost, 2026; Cockerill, 2024). The claim is not rejected by any single measurement but by the joint configuration; that is the under-fudgibility mechanism doing its work, and it is also why the diagnostic is hard to game: an actor wishing to defeat it must coordinate distortions across a census bureau, a health ministry's death records, foreign military-balance assessments, and historical calibration data that predate the conflict.

The same structure prices the common defences of the claim. Each rescues feasibility on one axis by conceding ground elsewhere: large wartime recruitment relaxes the manpower constraint, but concedes that the war generates fighters faster than it removes them; differential undercounting of combatant deaths in the MoH record relaxes the demographic constraint, but concedes that the headline death toll is too low; a much larger true toll relaxes the arithmetic, but concedes a war more lethal than the claim's proponents otherwise allow. The framework does not adjudicate among these defences; it computes what each costs in standard errors of the input it moves (Section 4) and forces the trade-off into the open.

Relation to convergent evidence. The convergence table (Table 4) is, to our knowledge, the first side-by-side placement of the demographic-decomposition literature, the named-combatant documentary record, and a generative spatial model on the same scale. We read the residual disagreement—roughly $q \in [2\%, 25\%]$—as an honest statement of what current public data cannot resolve: it is exactly the width of the exposure-calibration question plus the classification question (who counts as a combatant, on which the named-list evidence and demographic methods can legitimately differ). Narrowing it further requires either a defensible conflict-specific measurement of μ or transparent, auditable combatant lists, not another aggregation rule.

Limitations.

1. Class AM is treated as a single bucket; finer age partitions (18–29, 30–49, 50+) tighten the bounds when age-resolved data exist, and the calibration of μ from near-cutoff age bands (Frost, 2026) is itself subject to sampling noise in small historical samples.
2. The exposure range $[\mu_{\text{lo}}, \mu_{\text{hi}}]$ is imported from historical calibration; if the present conflict's exposure structure is genuinely outside that range, only the exposure-agnostic bound and

the manpower bound survive. We report both throughout for this reason.

3. The radius standardises by *sampling* error; systematic error in an anchor (e.g. selection in who gets recorded as dead) is addressed by re-computing under alternative anchors, not by the radius itself. Section 6 therefore reports all anchors, with the most claim-favourable one as primary.
4. The radius is conservative for non-Gaussian errors; a bootstrap or non-parametric posterior is preferred for small demographic samples.
5. Combatant status is a legal-institutional category, not a demographic one. The framework bounds the share of deaths attributable to members of armed groups as counted by manpower estimates; disputes about who should count as a combatant move M and the interpretation of named lists, and enter the diagnostic through those channels.
6. The framework does not identify indirect deaths from blockade, famine, or healthcare collapse; these require separate excess-mortality methodology (Checchi and Roberts, 2008; Burnham et al., 2006; Degomme and Guha-Sapir, 2010; Gómez-Ugarte et al., 2025) and enter only via the denominator D, which we vary as a sensitivity dimension.

Reproducibility. All bounds, radii, and table entries in this paper are generated by a validation script that recomputes them from the raw inputs and fails on any discrepancy. In addition, the algebraic core of the framework (Appendix A) is formalised and machine-checked in Lean 4 with mathlib, and the remaining results are verified symbolically and numerically by dedicated scripts. Code, the formal proofs, the spatial simulator, and all inputs are available in the companion repository.

References

James O. Berger and L. Mark Berliner. Robust Bayes and empirical Bayes analysis with ε-contaminated priors. *Annals of Statistics*, 14(2):461–486, 1986. doi: 10.1214/aos/1176349933.

Gilbert Burnham, Riyadh Lafta, Shannon Doocy, and Les Roberts. Mortality after the 2003 invasion of Iraq: A cross-sectional cluster sample survey. *The Lancet*, 368(9545):1421–1428, 2006. doi: 10.1016/S0140-6736(06)69491-9.

Francesco Checchi and Les Roberts. Documenting mortality in crises: What keeps us from doing better. *PLOS Medicine*, 5(7):e146, 2008. doi: 10.1371/journal.pmed.0050146.

Matthew Ghobrial Cockerill. Civilian casualties in Gaza: Israel's claims don't add up. Action on Armed Violence (AOAV), 2024. `https://aoav.org.uk/2024/casualties-in-gaza-israels-claims-of-50-combatant-deaths-dont-add-up-at-least-74-of-the-dead-are-civilians/`.

Olivier Degomme and Debarati Guha-Sapir. Patterns of mortality rates in Darfur conflict. *The Lancet*, 375(9711):294–300, 2010. doi: 10.1016/S0140-6736(09)61967-X.

Sergio DellaPergola and Mark Zlochin. Population representativeness and interviewer-level heterogeneity in the Gaza Mortality Survey. *Lancet Global Health*, 2026. doi: 10.1016/j.langlo.2026.103991. Correspondence.

Karl Frost. Why Israeli claims of low civilian-to-combatant harm in Gaza do not hold up. Action on Armed Violence (AOAV), 2026. `https://aoav.org.uk/2026/why-israeli-claims-of-low-civilian-to-combatant-harm-in-gaza-do-not-hold-up/`.

Philippe Gagnon. Robustness against conflicting prior information in regression. *Bayesian Analysis*, 2023. doi: 10.1214/22-BA1330. Advance publication.

Ana C. Gómez-Ugarte, Irena Chen, Enrique Acosta, Ugofilippo Basellini, and Diego Alburez-Gutierrez. Accounting for uncertainty in conflict mortality estimation: An application to the Gaza war in 2023–2024. *Population Health Metrics*, 23, 2025. doi: 10.1186/s12963-025-00422-9.

Peter J. Huber. Robust estimation of a location parameter. *Annals of Mathematical Statistics*, 35(1):73–101, 1964. doi: 10.1214/aoms/1177703732.

Peter J. Huber. *Robust Statistics*. Wiley, New York, 1981.

International Institute for Strategic Studies. The military balance 2024. Technical report, Routledge for IISS, 2024.

Israel Defense Forces. Public press briefings on combatant casualties, 2024–2026. As reported by Reuters and AP, 2024–2026.

Rasha Khatib, Martin McKee, and Salim Yusuf. Counting the dead in Gaza: Difficult but essential. *The Lancet*, 404(10449):237–238, 2024. doi: 10.1016/S0140-6736(24)01169-3.

Rasha Khatib et al. Traumatic injury mortality in the Gaza strip from Oct 7, 2023, to June 30, 2024: A capture–recapture analysis. *The Lancet*, 405, 2025. doi: 10.1016/S0140-6736(24)02678-3.

Quique Kierszenbaum, Yuval Abraham, et al. Revealed: Israeli military's own data indicates civilian death rate of 83% in Gaza war. The Guardian / +972 Magazine / Local Call joint investigation, 21 August 2025, 2025. `https://www.theguardian.com/world/ng-interactive/2025/aug/21/revealed-israeli-militarys-own-data-indicates-civilian-death-rate-of-83-in-gaza-war`.

Brendan Kline and Elie Tamer. Recent developments in partial identification. *Annual Review of Economics*, 15:125–150, 2023. doi: 10.1146/annurev-economics-051520-021124.

Yaniv Kubovich and Nir Hasson. IDF accepts Gaza Health Ministry death toll of over 71,000 Palestinians killed during the war. Haaretz, January 2026. URL `https://www.haaretz.com/israel-news/2026-01-29/ty-article/.premium/idf-accepts-gaza-health-ministry-estimate-of-over-70-000-palestinians-killed-in-the-war/0000019c-0918-dec4-adfd-fd5dde830000`.

Kristian Lum, Megan E. Price, and David Banks. Applications of multiple systems estimation in human rights research. *The American Statistician*, 67(4):191–200, 2013. doi: 10.1080/00031305.2013.821093.

Charles F. Manski. *Partial Identification of Probability Distributions*. Springer, New York, 2003.

Charles F. Manski. *Identification for Prediction and Decision*. Harvard University Press, Cambridge, MA, 2007.

Meir Amit Intelligence and Terrorism Information Center. Hamas deploys to recover and retain its military strength and influence over the Gaza Strip. Technical report, ITIC, February 2026. URL `https://www.terrorism-info.org.il/en/hamas-deploys-to-recover-and-retain-its-military-strength-and-influence-over-the-gaza-strip/`. Reports the post-ceasefire IDF assessment of roughly 20,000 operatives against more than 22,000 claimed killed, citing The Times of Israel, October 21, 2025.

Francesca Molinari. Microeconometrics with partial identification. In Steven N. Durlauf, Lars Peter Hansen, James J. Heckman, and Rosa L. Matzkin, editors, *Handbook of Econometrics, Vol. 7A*, pages 355–486. Elsevier, 2020.

Palestinian Central Bureau of Statistics. Gaza strip demographic and population releases. https://www.pcbs.gov.ps, 2023–2024.

Benjamin J. Radford, Yaoyao Dai, Niklas Stoehr, Aaron Schein, Mya Fernández, and Hanif Padilla. Estimating conflict losses and reporting biases. *Proceedings of the National Academy of Sciences*, 120(34):e2307372120, 2023. doi: 10.1073/pnas.2307372120.

Clionadh Raleigh, Andrew Linke, Håvard Hegre, and Joakim Karlsen. Introducing ACLED: An armed conflict location and event dataset. *Journal of Peace Research*, 47(5):651–660, 2010. doi: 10.1177/0022343310378914.

Reuters. Hamas has recruited 10,000–15,000 fighters since the war began, roughly matching its losses, US congressional sources say. Reuters, January 2025. URL https://www.reuters.com/world/middle-east/.

Joseph P. Romano and Azeem M. Shaikh. Inference for the identified set in partially identified econometric models. *Econometrica*, 78(1):169–211, 2010. doi: 10.3982/ECTA6706.

Royal United Services Institute. Hamas and Palestinian Islamic Jihad force-structure assessments. Technical report, RUSI, 2024.

Michael Spagat and Matthew Ghobrial Cockerill. The OHCHR report on Gaza: Insights, flaws, and the "Where's Daddy?" programme. Action on Armed Violence (AOAV), 2024. https://aoav.org.uk/2024/the-ohchr-report-on-gaza-insights-flaws-and-the-wheres-daddy-programme/.

Michael Spagat, Andrew Mack, Tara Cooper, and Joakim Kreutz. Estimating war deaths: An arena of contestation. *Journal of Conflict Resolution*, 53(6):934–950, 2009. doi: 10.1177/0022002709346253.

Michael Spagat, Jon Pedersen, Khalil Shikaki, Michael Robbins, Eran Bendavid, Håvard Hegre, and Debarati Guha-Sapir. Violent and non-violent death tolls for the Gaza conflict: new primary evidence from a population-representative field survey. *Lancet Global Health*, 14: e552–e559, 2026a. doi: 10.1016/S2214-109X(25)00522-4. Estimates 75,200 violent deaths (95% CI 63,600–86,800) to January 5, 2025; women, children, and people over 64 comprise 56.2% of violent deaths.

Michael Spagat, Jon Pedersen, Khalil Shikaki, Michael Robbins, Eran Bendavid, Håvard Hegre, and Debarati Guha-Sapir. Response to DellaPergola and Zlochin. *Lancet Global Health*, 2026b. doi: 10.1016/j.langlo.2026.103992. Correspondence.

Elie Tamer. Partial identification in econometrics. *Annual Review of Economics*, 2:167–195, 2010. doi: 10.1146/annurev.economics.050708.143401.

The Jerusalem Post. IDF in Gaza: Israel breaks through 10 out of 24 Hamas battalions. The Jerusalem Post, 2024. URL https://www.jpost.com/israel-news/defense-news/article-773066. First official IDF estimate of pre-war Hamas strength: 30,000 fighters in 24 battalions.

United Nations Office for the Coordination of Humanitarian Affairs. Reported impact snapshots, Gaza strip. https://www.ochaopt.org/data/casualties, 2025–2026.

United Nations Office of the High Commissioner for Human Rights. Six-month update report on the human rights situation in Gaza: 1 november 2023 – 30 april 2024. Technical report, OHCHR, 2024. With November 2024 thematic analysis of identified deaths.

Paola Vesco et al. The underreported death toll of wars: A probabilistic reassessment from a survey with UCDP coders. *Journal of Conflict Resolution*, 2026. doi: 10.1177/00220027251342000.

A Proofs

The identities, monotonicity statements, and optimality inequalities appearing in these proofs—together with the application arithmetic of Section 6, over exact rationals—have additionally been machine-checked in Lean 4 with mathlib; the development compiles with no unproven statements and is in the `formal/` directory of the companion repository, which documents exactly which statement corresponds to which theorem below. Steps involving posteriors, quantiles, and limits are formalised only in simplified kernel form there; for those, the proofs below are the complete argument, and they are additionally exercised by the symbolic and Monte-Carlo checks in the repository's verification scripts.

Proof of Theorem 3.3. Throughout, $w > 0$ and $f \le a$, so $w + \mu(a - f) > 0$ for every $\mu \ge 0$ and all denominators below are positive; the inputs (ω, w, a, f) are treated as population quantities (sampling error enters separately through the delta method of Section 3). Let λ be the civilian per-capita death hazard in the women-and-children class W, so class AM civilians have hazard $\mu\lambda$. Expected civilian deaths in W and in civilian-AM are $\lambda w N$ and $\lambda\mu(a - f)N$, so the share of *civilian* deaths landing in W is $w/(w + \mu(a - f))$. Combatant deaths land in AM, so only the civilian fraction $1 - q$ of deaths reaches W:

$$\omega = (1 - q)\,\frac{w}{w + \mu(a - f)},$$

which solves to (3); moreover $\partial_\mu q = -\omega(a - f)/w \le 0$, so $q(\mu)$ is nonincreasing (strictly decreasing unless $\omega = 0$ or $a = f$).

Attainability. Fix $\mu^* \in [\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}]$ with $q(\mu^*) \in (0, 1)$ and any $\lambda > 0$ small enough that the implied class death counts stay below the class populations. The configuration with combatant share $q(\mu^*)$ and hazards $(\lambda, \mu^*\lambda)$ reproduces the observed ω exactly, so $q(\mu^*)$ cannot be excluded by the data. Boundary values are attained in the closure: $q(\mu^*) = 1$ forces $\omega = 0$ (no civilian deaths, the limit $\lambda \downarrow 0$), and $q(\mu^*) = 0$ is attained directly with $\lambda > 0$ and no combatant deaths.

Sharpness. Conversely, any configuration consistent with the model has some $\mu \in [\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}]$, and the display then forces $q = q(\mu)$. Since $\mu \mapsto q(\mu)$ is continuous and monotone, the set of attainable values is exactly the interval $[q(\mu_{\mathrm{hi}}), q(\mu_{\mathrm{lo}})]$; intersecting with $[0, 1]$, the logical range of a share, gives the identified set. The benchmark identity (2) is the case $\mu = 1$, using $w + a = 1$. Corollary 3.4 follows since $D_M \le M$ and $D > 0$ give $q = D_M/D \le M/D$; the joint set is the intersection of two intervals and is empty exactly when the stated endpoints fail to be ordered. □

Proof of Proposition 2.1. A linear combination $\sum_i c_i\hat{\theta}_i$ of independent unbiased sources is unbiased iff $\sum_i c_i = 1$, and its variance is $\sum_i c_i^2\sigma_i^2$. By Cauchy–Schwarz,

$$1 = \Big(\sum_i c_i\Big)^2 = \Big(\sum_i (c_i\sigma_i)\cdot\sigma_i^{-1}\Big)^2 \le \Big(\sum_i c_i^2\sigma_i^2\Big)\Big(\sum_i \sigma_i^{-2}\Big),$$

so $\sum_i c_i^2\sigma_i^2 \geq \left(\sum_i \sigma_i^{-2}\right)^{-1} = \hat\sigma^2_{\text{agg}}$, with equality iff $c_i\sigma_i \propto \sigma_i^{-1}$, i.e. $c_i \propto \sigma_i^{-2}$—the weights in (1). The bias expression is linearity of expectation: $\mathbb{E}\sum_i c_i\hat\theta_i - \theta = \sum_i c_i b_i$ with $c_i = \hat\sigma^2_{\text{agg}}/\hat\sigma_i^2$.

For the contamination statement, write $\hat\theta_i = \theta + e_i + Z_i\Delta_i$, where e_i is the honest (mean-zero) error, $Z_i \in \{0,1\}$ indicates contamination ($\mathbb{E}Z_i \leq \varepsilon_i$), and Δ_i is the contaminating displacement, which by hypothesis satisfies $|\Delta_i| \leq R_i$ (the contaminating support lies in a set of diameter R_i containing the truth). Then

$$|\mathbb{E}\hat\theta_{\text{agg}} - \theta| = \Big|\sum_i c_i\,\mathbb{E}[Z_i\Delta_i]\Big| \leq \sum_i c_i\,\varepsilon_i R_i = \sum_i \varepsilon_i R_i\,\frac{\hat\sigma^2_{\text{agg}}}{\hat\sigma_i^2},$$

by the triangle inequality; no independence between the Z_i is needed. The bound controls the *expected* bias—the realised displacement on a contamination event can be as large as c_iR_i. ☐

Proof of Theorem 4.3. Write C for the constraint set of (5), the triples (ω', w', M') with $q^{(i)} \in \mathcal{F}((\omega', w', \hat a, \hat f, M', \hat D))$; if $C = \emptyset$ the infimum is $+\infty$ and both directions hold vacuously, so assume $C \neq \emptyset$. The objective is nonnegative, so $\rho^{(i)} \geq 0$ always.

($\Leftarrow$) If $q^{(i)} \in \mathcal{F}(\hat x)$, the point $(\hat\omega, \hat w, \hat M) \in C$ attains objective zero, so $\rho^{(i)} = 0$.

($\Rightarrow$) Suppose $\rho^{(i)} = 0$ and take $(\omega_n', w_n', M_n') \in C$ with objective $\to 0$. Because the M-penalty is one-sided, this yields $\omega_n' \to \hat\omega$, $w_n' \to \hat w$, and $(M_n' - \hat M)_+ \to 0$ only (*not* $M_n' \to \hat M$; M_n' may stay far below $\hat M$), so we pass to the limit through the constraints rather than through the minimiser. Each feasible point has some $\mu_n \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$ with $q^{(i)} = 1 - \omega_n'(w_n' + \mu_n(\hat a - \hat f))/w_n'$. Pass to a subsequence with $\mu_n \to \mu^*$ in the compact interval; since $\hat w > 0$ and $w_n' \to \hat w$, eventually $w_n' \geq \hat w/2 > 0$, so the map $(\omega', w', \mu) \mapsto 1 - \omega'(w' + \mu(\hat a - \hat f))/w'$ is continuous along the sequence and $q^{(i)} = 1 - \hat\omega(\hat w + \mu^*(\hat a - \hat f))/\hat w$: the exposure constraint holds at the observed $(\hat\omega, \hat w)$. For the manpower constraint, $q^{(i)}\hat D \leq M_n' \leq \hat M + (M_n' - \hat M)_+ \to \hat M$, so $q^{(i)}\hat D \leq \hat M$. Both membership conditions of (4) hold at $\hat x$, hence $q^{(i)} \in \mathcal{F}(\hat x)$.

For the second statement, any rationalisation $(\omega', w', M') \in C$ satisfies, because the infimum lower-bounds the objective on C, $\max\{|\omega' - \hat\omega|/\sigma_\omega,\ |w' - \hat w|/\sigma_w,\ (M' - \hat M)_+/\sigma_M\} \geq \rho^{(i)}$; a maximum of three terms is $\geq \rho^{(i)}$ only if at least one term is, so at least one penalised displacement is $\geq \rho^{(i)}$ standardised units. Downward moves of M' are unpenalised by construction and rightly so: they only shrink $\mathcal{F}$ (the constraint $q\hat D \leq M'$ tightens), so they can never rescue a claim. ☐

Proof of Proposition 5.1. (i) Condition on the realised pattern T. The sources outside T are honest, and by the proposition's standing hypothesis every posterior in play is supported on the identified set implied by the sources it uses; a posterior formed from the honest sources plus arbitrary reports from T is therefore supported on the identified set with the T-sources removed. Removing sources removes constraints, so this removal set contains the face-value identified set—in particular the face-value posterior support, and with it both L_0 and U_0—and it has diameter R_T by definition. Any two points of a set of diameter R_T are within R_T of each other, so each endpoint of a support-respecting credible interval (both its endpoints lie in the posterior support, hence in the removal set) is within R_T of the corresponding endpoint of I_0.

(ii) The contamination indicators are independent, so pattern T realises with probability $P(T) = \prod_{i\in T}\varepsilon_i \prod_{i\notin T}(1 - \varepsilon_i)$, and by part (i) the endpoint displacement on that event is at most R_T ($R_\emptyset = 0$). Hence the expected displacement is at most $\sum_T P(T)R_T$. Splitting off the singleton patterns and bounding $\prod_{j\neq i}(1 - \varepsilon_j) \leq 1$,

$$\sum_T P(T)R_T \ \leq\ \sum_i \varepsilon_i R_i + \sum_{|T|\geq 2} P(T)R_T,$$

and $P(T) \leq \varepsilon_{\max}^{|T|}$ makes the second sum $O(\varepsilon^2)$ (it vanishes identically when at most one source can be contaminated). Note that R_T for $|T| \geq 2$ is *not* bounded by $\sum_{i\in T} R_i$ in general (two mutually corroborating sources can each have a small removal diameter while their joint removal is much larger), which is why (6) is stated as a first-order band rather than a uniform envelope.

(iii) Write the contaminated posterior as the mixture $\hat{F}_\varepsilon = (1-\pi)\hat{F}_0 + \pi G$, where G is the (sub-)posterior on contaminated components and $\pi \le \tau$ by assumption. Since $|\hat{F}_0(x) - G(x)| \le 1$ for all x, $\sup_x |\hat{F}_\varepsilon(x) - \hat{F}_0(x)| \le \pi \le \tau$. Let $u \in \{\alpha/2, 1-\alpha/2\}$ and abbreviate $Q_0 = \hat{Q}_0(u)$, $Q_\varepsilon = \hat{Q}_\varepsilon(u)$, quantiles taken as generalised inverses. $\hat{F}_0$ is continuous on the segment in question (it has a density there), so $\hat{F}_0(Q_0) = u$, and the sup-bound $|\hat{F}_\varepsilon - \hat{F}_0| \le \tau$ holds at left limits as well. The generalised inverse satisfies $\hat{F}_\varepsilon(Q_\varepsilon^-) \le u \le \hat{F}_\varepsilon(Q_\varepsilon)$, whence

$$\hat{F}_0(Q_\varepsilon) - \tau \le \hat{F}_\varepsilon(Q_\varepsilon^-) \le u \le \hat{F}_\varepsilon(Q_\varepsilon) \le \hat{F}_0(Q_\varepsilon) + \tau,$$

i.e. $|\hat{F}_0(Q_\varepsilon) - \hat{F}_0(Q_0)| = |\hat{F}_0(Q_\varepsilon) - u| \le \tau$, and since $\hat{F}_0$ has density $\ge m$ on the segment between Q_0 and Q_ε, the left side is $\ge m\,|Q_\varepsilon - Q_0|$; rearranging gives $|Q_\varepsilon - Q_0| \le m^{-1}\tau$. The weight condition is substantive, not automatic: posterior mixture weights are proportional to $P(T)$ times the component marginal likelihoods, so an adversarial component that fits the observed data *better* than the clean model can carry posterior weight exceeding its prior probability τ. □

B The spatial Bayesian model

This appendix documents the forward model, priors, likelihood, and inference behind the posterior reported in Section 6, Step 5. Code and inputs are in the `gaza_sim/` directory of the companion repository, and the posterior is reproducible from a fixed random seed. We emphasise its role: it is *one* model-based point within the assumption-free identified set, and its low posterior q reflects its priors and structural choices as much as the data. Readers who reject those choices should fall back to the bounds, which do not depend on them.

Geometry and population. Gaza's five governorates are split into $C = 400$ equal-population cells (80 per governorate), with cell populations N_c from the PCBS 2023 total. Demographic fractions (women-and-children share w, adult-male share a) are held constant across cells at the Gaza-wide values.

Latent parameters and priors. Each posterior draw samples, independently and uniformly on the stated ranges:

Parameter	Range	Meaning
M	$[35\text{k}, 60\text{k}]$	total militant stock
γ	$[0, 1.5]$	spatial clustering amplification of militants
σ	$[0.3, 1.2]$	SD of the cell-level latent log-density
α	$[0, 8]$	strike-targeting concentration on militant density
β	$[0.6, 1.6]$	targeting non-linearity exponent
μ_M	$[1, 2.5]$	civilian adult-male exposure multiplier
ϵ_C	$[0.7, 1]$	child exposure down-weight
$\bar{d}$	$[1.2, 2.8]$	expected kills per strike
K	$[29\text{k}, 40\text{k}]$	total air-to-ground strikes
u	$[1/1.7, 1]$	recovery factor (MoH observed / true direct)

Militants are allocated across cells proportionally to $N_c \exp(\gamma u_c)$ with $u_c \sim \mathcal{N}(0, \sigma^2)$, capped at 30% of each cell's adult-male population and rescaled to the total M.

Strike-kill model. Strikes are allocated across cells with weight $N_c(1 + \alpha m_c/N_c)^\beta$, where m_c is the militant count in cell c. Within a cell, a death is a militant, a civilian adult man, or a woman/child with probabilities proportional to m_c, $(N_c a - m_c)\mu_M$, and $N_c w\,\epsilon_C$ respectively. Expected class-specific deaths are summed over cells; observed deaths are true direct deaths times the recovery factor u.

Likelihood. Three observables: (i) the MoH cumulative total (log-Gaussian, 7% scale); (ii) the adult-male share among the dead against the OHCHR verified sample (Beta likelihood at its sample size, $n = 8{,}119$); and (iii) the same share against the MoH full record, down-weighted to an effective sample size of $n/5$ to reflect cruder demographic categorisation. The two demographic likelihoods enter with weight 1/2 each.

Inference. Importance sampling with 2×10^5 prior draws; the effective sample size and all posterior quantiles are computed under the normalised weights. The reported posterior is $q = 2.5\%$ (95% credible interval $[1.6\%, 3.8\%]$).

Sensitivities and limitations. The posterior concentrates at low q for two structural reasons that a critic can legitimately contest: the exposure multiplier prior is capped at $\mu_M \leq 2.5$ (values above the cap would push q lower still, but a mass of prior near $\mu_M = 1$ pulls the posterior up much less than the OHCHR-weighted demographic likelihood pulls it down), and the OHCHR sample—which over-represents residential-strike deaths where women and children predominate (Spagat and Cockerill, 2024)—enters the likelihood at full sample weight. Re-anchoring the demographic likelihood entirely on the MoH record moves the posterior median up but keeps it far inside the identified set. Unlike the exposure calibration of Section 3, the model has not been validated on historical conflicts with known combatant status. Because these choices are contestable, the paper's claims do not rest on this posterior; it appears as one row of Table 4.

C The 81-conflict dataset

Table 5 lists every conflict in the dataset behind the overview of Section 7, with midpoint death totals and the implied civilian-share intervals. The table is generated by the same script that draws Figures 3 and 4; per-conflict source notes, the indirect-death decomposition, and the accounting rules are documented in the supplement.

Table 5. The 81-conflict dataset behind Figures 3 and 4 (conflicts with at least 1,000 attributed deaths shown). Deaths are midpoints of the source ranges (direct military + civilian, including indirect where sources attribute it); the civilian-share interval spans the most and least civilian-heavy readings of the source ranges. Per-conflict sources are in the supplement.

Conflict	Period	Deaths (mid)	Civilian share	lo–hi
Boxer Rebellion	1899–1901	47k	75%	63–92%
Philippine-American War	1899–1902	211k	89%	86–92%
Herero and Namaqua Genocide	1904–1908	82.5k	98%	97–99%
Russo-Japanese War	1904–1905	187k	21%	19–24%
Mexican Revolution	1910–1920	1.76M	89%	73–95%
First and Second Balkan Wars	1912–1913	326k	32%	18–43%
Armenian Genocide and late Ottoman genocide...	1914–1923	2.54M	100%	100–100%
World War I	1914–1918	15.9M	40%	31–47%
Russian Civil War	1917–1923	7.32M	86%	41–97%

Table 5 continued

Conflict	Period	Deaths (mid)	Civilian share	lo–hi
Greco-Turkish War and population exchange	1919–1922	1.3M	96%	94–98%
Irish War of Independence and Civil War	1919–1923	3.49k	23%	19–27%
Rif War	1921–1927	71.6k	17%	11–31%
Chaco War	1932–1935	94.5k	2%	1–4%
Soviet repression: Holodomor and Great Purge	1932–1939	14M	100%	100–100%
The Holocaust	1933–1945	19.3M	83%	81–84%
Second Italo-Ethiopian War	1935–1937	609k	62%	49–84%
Spanish Civil War	1936–1939	356k	51%	38–61%
Second Sino-Japanese War	1937–1945	13.6M	74%	62–83%
World War II - European/African/Atlantic Th...	1939–1945	48.8M	61%	56–66%
World War II - Pacific Theater/Asia-Pacific...	1941–1945	25.4M	77%	67–83%
Bengal Famine of 1943	1943–1944	2.65M	100%	100–100%
Chinese Civil War	1945–1949	6.07M	86%	86–86%
Indonesian National Revolution	1945–1949	164k	48%	19–72%
First Indochina War	1946–1954	631k	44%	27–60%
1948 Arab-Israeli War/Nakba	1947–1949	24.2k	52%	42–62%
Partition of India	1947–1948	700k	100%	100–100%
Malayan Emergency	1948–1960	11.4k	25%	22–28%
Korean War	1950–1953	1.56M	39%	31–49%
Mau Mau Uprising	1952–1960	63.4k	65%	45–83%
Algerian War of Independence	1954–1962	577k	70%	63–75%
Vietnam War	1955–1975	2.04M	38%	23–51%
Great Leap Forward famine	1958–1962	30M	100%	100–100%
Congo Crisis	1960–1965	12.1k	60%	59–60%
Guatemalan Civil War	1960–1996	164k	100%	100–100%
North Yemen Civil War	1962–1970	66.2k	0%	0–0%
Indonesian mass killings of 1965-66	1965–1966	600k	100%	100–100%
Nigerian Civil War	1967–1970	1.25M	100%	100–100%
Six-Day War	1967–1967	15.7k	0%	0–0%
Bangladesh Liberation War/1971 genocide	1971–1971	2.18M	99%	89–100%
Yom Kippur War	1973–1973	16k	1%	0–2%
Ethiopian Civil War/Derg/1983-85 famine	1974–1991	596k	83%	81–85%
Angolan Civil War	1975–2002	70.6k	96%	95–96%
Indonesian occupation of East Timor	1975–1999	150k	98%	96–98%
Khmer Rouge Cambodian Genocide	1975–1979	2.03M	99%	98–99%
Lebanese Civil War	1975–1990	22.5k	95%	88–97%
Mozambican Civil War	1977–1992	50.7k	100%	100–100%
Cambodian-Vietnamese War	1978–1989	588k	89%	83–94%
Salvadoran Civil War	1979–1992	93.3k	70%	61–77%
Soviet-Afghan War	1979–1989	115k	0%	0–0%
Iran-Iraq War	1980–1988	470k	24%	17–33%
Second Sudanese Civil War	1983–2005	27k	100%	100–100%
Sri Lankan Civil War	1983–2009	95.5k	45%	42–48%
First Intifada	1987–1993	2.21k	97%	97–97%
Nagorno-Karabakh wars	1988–2023	36.2k	5%	4–7%
Gulf War	1990–1991	209k	66%	41–90%
Rwandan Genocide and Rwandan Civil War	1990–1994	822k	100%	100–100%
Somali Civil War	1991–	563k	99%	99–100%
Yugoslav Wars - Bosnia/Croatia	1991–1995	71.6k	1%	0–1%

Table 5 continued

Conflict	Period	Deaths (mid)	Civilian share	lo–hi
First Chechen War	1994–1996	56.3k	71%	56–86%
First Congo War	1996–1997	151k	100%	100–100%
Kosovo War	1998–1999	11.8k	73%	71–75%
Second Congo War	1998–2003	3.92M	100%	97–100%
Second Chechen War	1999–2009	56.2k	67%	48–82%
Second Intifada	2000–2005	4.6k	65%	53–75%
War in Afghanistan	2001–2021	183k	24%	19–29%
Darfur conflict/genocide	2003–2020	282k	100%	100–100%
Iraq War	2003–2011	391k	88%	87–88%
Mexican Drug War	2006–	6.45k	3%	2–3%
Boko Haram insurgency/Lake Chad conflict	2009–	370k	94%	94–94%
Libyan Civil Wars	2011–2020	10.4k	10%	7–12%
Syrian Civil War	2011–2024	727k	53%	46–59%
South Sudanese Civil War	2013–2020	1.15k	100%	100–100%
War against ISIS in Iraq and Syria	2014–2019	145k	30%	16–47%
War in Donbas	2014–2022	11.6k	6%	6–7%
Yemeni Civil War	2014–	11.2k	82%	78–85%
Rohingya genocide/Myanmar military operatio...	2016–	22.4k	98%	96–99%
Tigray War	2020–2022	492k	100%	100–100%
Myanmar civil war	2021–	17.3k	71%	71–71%
Russian invasion of Ukraine	2022–	490k	6%	4–10%
Israel-Hamas/Gaza war	2023–	93.3k	96%	93–98%
Sudan war	2023–	20.1k	100%	100–100%